

4. CSS Selectors

Selectors tell CSS **which HTML elements you want to style**.

The most important selectors are:

1. Element selector
2. ID selector
3. Class selector
4. Universal selector
5. Group selector

4.1 Element Selector

Selects HTML elements by their tag name.

```
p {  
    color: blue;  
}
```

This applies to **all <p> elements**.

```
<p>First paragraph</p>  
<p>Second paragraph</p>  
<p>Third paragraph</p>
```

All three become blue.

5. ID Selector

ID is selected using #.

HTML:

```
<p id="para1">Hello</p>
```

CSS:

```
#para1 {  
    color: red;  
}
```

Syntax:

```
#idName {  
  property: value;  
}
```

An ID should normally be **unique** on a page.

6. Class Selector

Class is selected using .(dot)

HTML:

```
<p class="important">Hello</p>  
<p class="important">Welcome</p>
```

CSS:

```
.important {  
  color: red;  
}
```

Both paragraphs become red.

Classes are extremely important because the **same class can be used on multiple elements**.

7. Universal Selector

The universal selector is: *

It selects **all elements**.

Example:

```
* {  
  margin: 0;  
  padding: 0;  
}
```

This removes the default margin and padding from all elements.

8. Group Selector

You can style multiple elements together.

Instead of:

```
h1 {  
  color: blue;  
}  
  
h2 {  
  color: blue;  
}  
  
p {  
  color: blue;  
}
```

You can write:

```
h1, h2, p {  
  color: blue;  
}
```

This is called a **group selector**.

9. Colors

CSS allows you to set colors in several ways.

Color name

```
h1 {  
  color: red;  
}
```

Hexadecimal

```
h1 {  
  color: #ff0000;  
}
```

RGB

```
h1 {  
  color: rgb(255, 0, 0);  
}
```

RGBA

```
h1 {  
  color: rgba(255, 0, 0, 0.5);  
}
```

The last value represents **opacity**.

10. Background Color

Use:

background-color

Example:

```
body {  
    background-color: lightblue;  
}
```

Or:

```
div {  
    background-color: yellow;  
}
```

11. Text Properties

Some important text properties are:

color

```
p {  
    color: red;  
}
```

text-align

```
h1 {  
    text-align: center;  
}
```

Possible values:

```
left  
center  
right  
justify
```

text-decoration

```
a {  
    text-decoration: none;  
}
```

Other values:

underline
overline
line-through
none

text-transform

```
p {  
    text-transform: uppercase;  
}
```

Other values:

uppercase
lowercase
capitalize
none

12. Fonts

Font family

```
p {  
    font-family: Arial;  
}
```

You can provide multiple fonts:

```
p {  
    font-family: Arial, sans-serif;  
}
```

If Arial isn't available, the browser uses `sans-serif`.

Font size

```
p {  
    font-size: 20px;  
}
```

Common units:

px
%
em
rem
vh
vw

We'll learn these properly later.

Font weight

```
p {  
    font-weight: bold;  
}
```

Or:

```
p {  
    font-weight: 700;  
}
```

Font style

```
p {  
    font-style: italic;  
}
```

13. CSS Units

CSS commonly uses:

px

```
font-size: 20px;
```

Fixed pixel size.

%

```
width: 50%;
```

Relative to the parent/container.

em

```
font-size: 2em;
```

Relative to the element's font size/context.

rem

```
font-size: 2rem;
```

Relative to the root (html) font size.

vh

Viewport height:

```
height: 50vh;
```

vw

Viewport width:

```
width: 50vw;
```

14. Width and Height

```
div {  
  width: 300px;  
  height: 200px;  
}
```

You can also use percentages:

```
div {  
  width: 50%;  
}
```

15. Borders

Basic border:

```
div {  
  border: 2px solid black;  
}
```

Structure:

```
border-width  
border-style  
border-color
```

For example:

```
div {  
  border-width: 2px;  
  border-style: solid;  
  border-color: black;  
}
```

Common border styles:

solid
dashed
dotted
double
none

16. Border Radius

Used to make corners rounded.

```
div {  
    border-radius: 10px;  
}
```

For a circle:

```
img {  
    border-radius: 50%;  
}
```

17. Margin

Margin = space outside the element.

```
div {  
    margin: 20px;  
}
```

You can also specify individual sides:

```
div {  
    margin-top: 10px;  
    margin-right: 20px;  
    margin-bottom: 10px;  
    margin-left: 20px;  
}
```

18. Padding

Padding = space inside the element, between content and border.

```
div {  
    padding: 20px;  
}
```

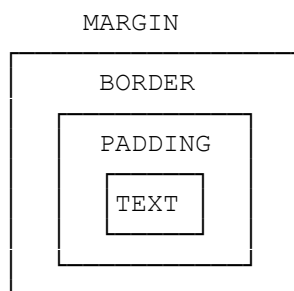
Example:

```
<div class="box">
```

```
    Hello
</div>
.box {
  border: 2px solid black;
  padding: 20px;
}
```

19. Margin vs Padding

This is VERY important.

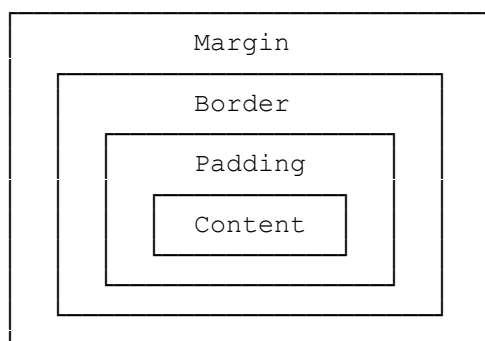


Margin → outside

Padding → inside

20. The CSS Box Model

Every HTML element is basically treated as a box:



The box model consists of:

Content → **Padding** → **Border** → **Margin**

This is one of the most important concepts in CSS.

21. Display Property

The `display` property controls how an element behaves.

Common values:

```
display: block;  
display: inline;  
display: inline-block;  
display: none;
```

Block

```
<div>Hello</div>
```

A block element generally takes the full available width.

Examples:

```
div  
p  
h1  
section  
header
```

Inline

Examples:

```
span  
a  
strong  
em
```

Inline elements generally occupy only the space they need.

22. Flexbox

Flexbox is used to arrange elements.

Example:

```
.container {  
    display: flex;  
}
```

Then:

```
<div class="container">
```

```
<div>Box 1</div>
<div>Box 2</div>
<div>Box 3</div>
</div>
```

Important Flexbox properties:

```
display: flex;
flex-direction: row;
justify-content: center;
align-items: center;
gap: 20px;
```

Example:

```
.container {
  display: flex;
  justify-content: center;
  align-items: center;
  gap: 20px;
}
```

We'll cover Flexbox separately in depth.

23. CSS Comments

Comments are written as:

```
/* This is a CSS comment */
```

Example:

```
/* Heading styling */
h1 {
  color: blue;
}
```

24. A Complete Beginner Example

HTML

```
<!DOCTYPE html>
<html>
<head>
  <title>My First CSS</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>
```

```
<h1>Welcome to CSS</h1>

<p class="text">
  This is my first CSS webpage.
</p>

<button>Click Me</button>

</body>
</html>
```

style.css

```
body {
  background-color: lightblue;
}

h1 {
  color: darkblue;
  text-align: center;
  font-size: 40px;
}

.text {
  color: black;
  font-size: 20px;
  text-align: center;
}

button {
  background-color: blue;
  color: white;
  padding: 10px 20px;
  border: none;
  border-radius: 5px;
}
```

CSS Learning Order

If you're learning CSS from scratch, I recommend this order:

1. What is CSS
- ↓
2. CSS Syntax
- ↓
3. Inline / Internal / External CSS
- ↓
4. Selectors
- ↓
5. Colors & Backgrounds
- ↓
6. Text & Fonts
- ↓
7. Width & Height
- ↓
8. Borders
- ↓

9. Margin & Padding
↓
10. Box Model
↓
11. Display
↓
12. Position
↓
13. Flexbox
↓
14. Grid
↓
15. Pseudo-classes
↓
16. Pseudo-elements
↓
17. Transitions
↓
18. Transform
↓
19. Animations
↓
20. Media Queries / Responsive Design
↓
21. Advanced CSS